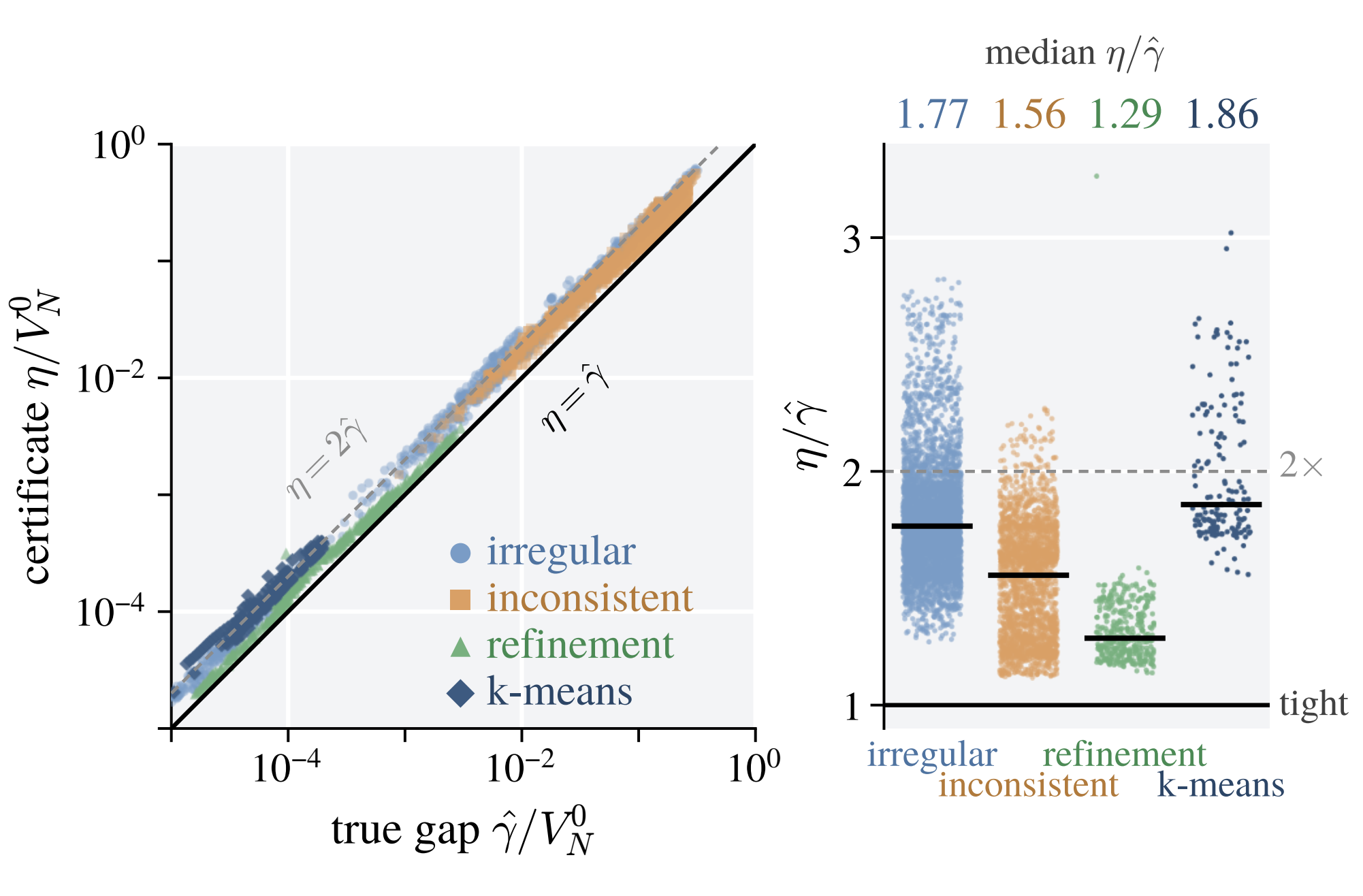

median $\eta/\hat{\gamma}$
1.77
1.56
1.29
1.86
10^0
10^{-2}
10^{-4}
certificate η/V_N^0
$\eta = 2\hat{\gamma}$
$\eta = \hat{\gamma}$
irregular
inconsistent
refinement
k-means
10^{-4}
10^{-2}
10^0
true gap $\hat{\gamma}/V_N^0$
3
2
1
$\eta/\hat{\gamma}$
2×
tight
irregular
refinement
inconsistent
k-means